

Semesterprogram:
w.BA.XX.3DA-WIN.XX

Fall Semester 2026

2x14x2 lessons

General information:

Lecturers:

- Dr. Mario Gellrich (module responsible)
- Dr. Alexander Walter

Documents:

- All course materials can be found on the electronic learning platform "Moodle" and on GitHub.

Tools and software:

- Generally, we will use GitHub Codespaces, Python and Jupyter notebooks for all exercises.
- Installation instructions of additional tools will be provided or linked on Moodle.

Location/teaching format:

- **Large classes (german: GK) are organized as follows:**
 - All lectures will be held on-site in Winterthur (see WebUntis for rooms and schedules).
- **Small classes (german: KK) are organized as follows:**
 - In the first week of the semester, all small classes will be held on-site in Winterthur (see WebUntis for rooms and schedules).
 - From the second week of the semester onwards, small classes will take place online ([MS-Teams](#)).
 - Note that all online small classes can be attended by every student at any time!
- Changes in the location and form of instruction will be announced in a timely manner.

Weekly exercises:

The weekly exercises in the small classes must not be submitted on Moodle; they serve as preparation for and will be part of the written exam.

End of module exam / final grade:

- Only the current module description is relevant with regard to the exam and composition of final grade.
- The final grade is determined as:
 - **60% grade of written exam (regular end-of-module exam, MEP)**
 - **40% grade of project work with the two parts (each 20%):**
 - **the final project submission on Moodle**
 - **the group presentation of the project concept on-site in Winterthur**

- The **on-site project presentation** of the **project concept** covers the background, problem statement, objectives, research question, data, methods, and expected results (not the finished project). For detailed requirements see Moodle. They take place at the beginning of the lectures (large classes, on-site in Winterthur) from week 10 onwards (3-4 groups per lecture, 10 min. per group).

- Date and time for the presentation of the project concept can be found at the beginning of the semester under the link to register for the project work on Moodle. Note that the lecturer will define the date/time once all groups are registered by students.

The written exam will ...

- last 60 minutes
- be open book
- cover the entire material of the module
- contain single-choice questions (4 possible answers per question, only one is correct).

Submission of project work:

Deadline for the submission of the final project is January 10, 2027 (11:59 p.m.)

Course materials (literature basis of the course; not mandatory but recommended)	
Source (1): Navlani, A., Fandango, A., & Idris, I. (2021). Python Data Analysis: Perform data collection, data processing, wrangling, visualization, and model building using Python. Packt Publishing Ltd.	
Source (2): Sundnes, J. (2020). Introduction to Scientific Programming with Python (p. 148). Springer Nature.	
Source (3): Stratis, K. (2020). Use Sentiment Analysis With Python to Classify Movie Reviews.	
Source (4): Geeks for Geeks Python Tutorial (n.d.). https://www.geeksforgeeks.org/python/python-programming-language-tutorial/	
Source (5): Real Python Courses (n.d.). https://realpython.com/search?kind=course&order=newest	
Source (6): w3schools Python Tutorial (n.d.). https://www.w3schools.com/python	
Source (7): Pandas User Guide (n.d.) https://pandas.pydata.org/docs/user_guide/index.html	
Source (8): Mitchell, R. (2024). Web Scraping with Python: Data Extraction from the Modern Web. 3rd edition. O'Reilly Media	
Source (9): Geopandas Documentation (n.d.). https://geopandas.org/en/stable/docs.html	
Source (10): Denis, D.J. (2021). Applied univariate, bivariate, and multivariate statistics using Python. John Wiley & Sons	
Source (11): Samaddar, S., & Nargundkar, S. (Eds.). (2019). Data analytics: Effective methods for presenting results. CRC Press.	
Source (12): McKinney, W. (2022). Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter. 3rd edition. O'Reilly Media. Open access: https://wesmckinney.com/book	
Source (13): Python AI: How to Build a Neural Network & Make Predictions. https://realpython.com/python-ai-neural-network	

Week	Main topics	Sources	Assignments
Week 1 (15.09. - 18.09.)	Introduction to Data Analytics, Categorization of Data, Data Sources & Data Collection Methods	Src.: 1,2,5,8	Exercises with Python according to the content of lessons
Week 2 (21.09. - 25.09.)	Data Types, Data Structures File Formats, Character Encoding, Import and Export of Data in Python	Src.: 1,2,4,5,6	Exercises with Python according to the content of lessons
Week 3 (28.09. - 02.10.)	Data Preparation Combining & Organizing Data	Src.: 1,5,7	Exercises with Python according to the content of lessons
Week 4 (05.10. - 09.10.)	Exploratory Data Analysis (EDA)	Src.: 1,2,5,7	Exercises with Python according to the content of lessons
Week 5 (12.10. - 16.10.)	Representation & Analysis of Geographical Data	Src.: 9	Exercises with Python according to the content of lessons
Week 6 (19.10. - 23.10.)	Statistical Hypothesis Testing (Part I)	Src.: 10	Exercises with Python according to the content of lessons
Week 7 (26.10.-30.10.)	Statistical Hypothesis Testing (Part II)	Src.: 1,10	Exercises with Python according to the content of lessons

Week	Main topics	Sources	Assignments
Week 8 (02.11. - 06.11.)	Regression Analysis (OLS, RT)	Src.: 1,10	Exercises with Python according to the content of lessons
Week 9 (09.11. - 13.11.)	Classification Analysis (CT, RF)	Src.: 1,10	Exercises with Python according to the content of lessons
Week 10 (16.11. - 20.11.)	Cluster Analysis (k-means) <i>Group presentations: Project Concept</i>	Src.: 1,10	Exercises with Python according to the content of lessons
Week 11 (23.11. - 27.11.)	Text Mining & Sentiment Analysis <i>Group presentations: Project Concept</i>	Src.: 1,3,5	Exercises with Python according to the content of lessons
Week 12 (30.11. - 04.12.)	Introduction to Neural Networks <i>Group presentations: Project Concept</i>	Src.: 13	Exercises with Python according to the content of lessons
Week 13 (07.12. - 11.12.)	Presenting and Sharing Data Analysis Results <i>Group presentations: Project Concept</i>	Src.: 11	-
Week 14 (14.12. - 18.12.)	<i>Group presentations: Project Concept</i>	-	-
21.12. - 17.01.	Semester break	-	-
10.01.2027	Deadline project submission	-	-
	End-of-module exams	-	-

Workload Details

Study format	Workload	Hours
Contact hours	Contact hours include the lessons on the course days. Basic knowledge is acquired in the process.	56h
Guided Self-study	Guided self-study is a form of study that is organized and carried out independently by the students but is accompanied or guided by the lecturers. The lecturers are not present. In this course, this includes solving assignments independently.	56h

Independent selfstudy	Independent self-study is a form of study in which students work independently within the framework of the compulsory curriculum and are not accompanied by lecturers. In this course, this includes preparing for and following up lessons, studying literature and preparing for exams.	68h
	Total workload in hours	180h
	Total ECTS	6